NEXUS 2.0

TITLE PAGE

Team Name: Neural Nomads

Problem Statement: AI-Powered Medical Billing Auditor & Insurance Appeal Engine for India

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THE PROBLEM

The Healthcare Financing Crisis in India

Opaque. Complex. Exploitative.

Healthcare billing and insurance systems in India lack transparency, leading to **unchecked overcharges and denied claims.**

A patient in Pune was billed ₹2,200 for a procedure priced at ₹800 under CGHS norms — and had no way to know.

50Cr
Insured Population

8M
Annual Claim Denials

67%
Never Appealed

₹8K-15K
Average Overcharge

Why This Gap Exists?

- Non-standardized and highly complex hospital billing formats that vary between institutions
- Lack of accessible interpretation of IRDAI regulations and compliance requirements
- Absence of automated tools that can analyze bills and generate evidence-based appeals

Our Solution

What We Do: AI system that audits bills and auto-generates insurance appeals.

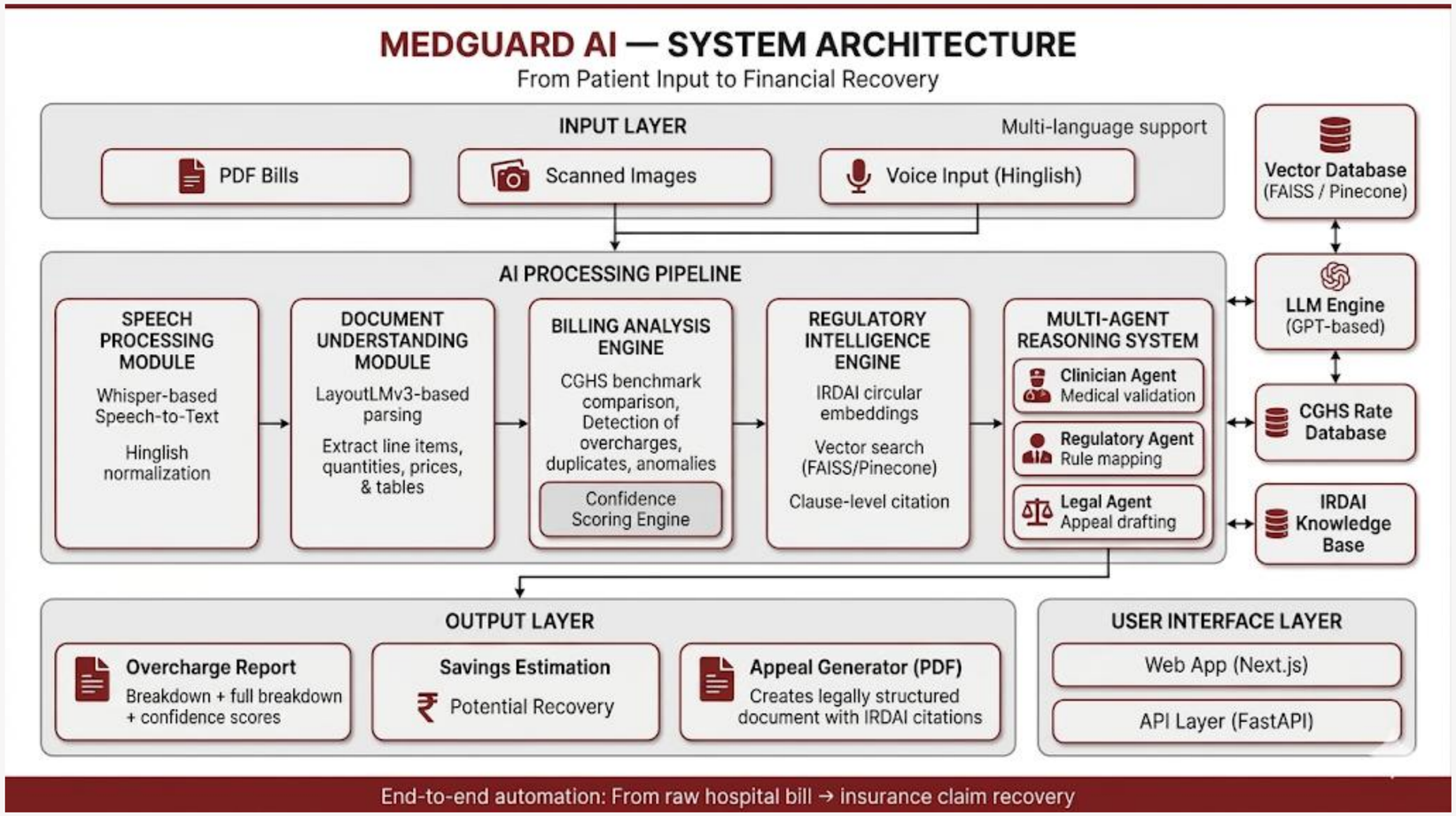
MedGuard AI uploads your hospital bill, flags every overcharge against CGHS rates, and auto-generates a legally grounded IRDAI-compliant appeal letter — in under 15 seconds.

No tool exists in India that does thisa Until now.

TECHNICAL APPROACH

Tech Stack Overview:

Layer	Technologies
Frontend	Next.js, Tailwind, ShadCN
Backend & APIs	FastAPI, API Gateway, Celery, Redis
AI Core	Whisper, LayoutLMv3, GPT Models
Data Layer	PostgreSQL, S3, FAISS / Pinecone
Output	ReportLab, Jinja2
Infrastructure & DevOps	Docker, Vercel, AWS, GitHub Actions, Razorpay



FEASIBILITY & VIABILITY

Feasibility:

Aspect	Key Points
Proven Technology Stack	Whisper, LayoutLMv3, GPT — production-ready
Data Availability	CGHS rates + IRDAI regulations (public, updatable)
Build Status	AdvocAI appeal pipeline live — ClaimShield layer in active development
Processing Performance	**~15s processing**, cloud-native
System Accessibility	PDFs, images, Hinglish voice input
Risk & Mitigation	Extend CGHS with NABH/private benchmarks

Viability:

Aspect	Key Points
Market Opportunity	**50Cr** insured, **8M** denials, **67%** unchallenged
Revenue Model	**₹99/appeal** + SaaS + API integrations
User Value Proposition	**₹8K–₹15K** savings per bill
Market Capture Potential	Even **0.1%** → **5K+** cases/year
Competitive Advantage	CGHS + IRDAI + Hinglish + automated appeals
Scalability & Expansion	Fraud detection, compliance, claims automation
Sustainability	Periodic IRDAI + CGHS updates

Built on public government data. Cloud-native. Zero dependency on proprietary hospital systems.

Phase 1 live: May 2, 2026 | GitHub: github.com/Viraj281105/Advocai

IMPACT & BENEFITS

Transforming Healthcare Financial Protection

What changes for 50 crore Indians:

MedGuard AI bridges the gap between complex healthcare billing and patient financial protection—enabling fast, accurate, and accessible recovery of unjustified charges.

⏱ 15 seconds. From raw hospital bill to legally grounded appeal letter.

1.

Accessibility First

Hinglish voice input + English/Hindi text — covers 600M+ Hindi speakers with zero technical barrier

2.

Accuracy Through AI

87%+ confidence scoring on every flagged overcharge — not just a flag, but a probability

3.

Speed at Scale

~15s processing vs hours/days → instant actionable insights vs. 3–7 days with a medical lawyer costing ₹3,000–₹8,000

4.

Savings Visibility

Average ₹12,400 potential recovery surfaced per bill — shown instantly, not buried in a report

67% of patients never appeal. MedGuard AI makes that number zero.

References

Source	Category	Role in MedGuard AI	Link
CGHS (Govt. of India)	Pricing Benchmark	Standard rates used to detect hospital overcharges	https://cghs.mohfw.gov.in
IRDAI	Insurance Regulation	Defines claim rules, approvals, and appeal frameworks	https://www.irdai.gov.in
IRDAI Annual Reports	Market Data	Provides claim denial rates and insurance trends	https://irdai.gov.in/annual-reports
National Health Authority (Ayushman Bharat)	Public Coverage Data	Validates scale of insured population in India	https://nha.gov.in
NDHM (Digital Health Mission)	Health Infrastructure	Supports digital health ecosystem & interoperability	https://ndhm.gov.in
NABH	Hospital Standards	Reference for private hospital practices (future benchmarking)	https://nabh.co
WHO (Health Financing)	Global Context	Validates need for transparency in healthcare financing	https://www.who.int/health-topics/health-financing
World Bank Health Data	Economic Validation	Supports macro-level healthcare spending insights	https://data.worldbank.org